

Energy Internet in China

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This section introduces Energy Internet (EI) in China and consists of three parts: concept and characteristics, key technologies, and representative demonstration projects. We first introduce the concept of EI, followed by six of its features: (1) Synergy of energy, (2) Higher energy efficiency, (3) Commercialization of energy, (4) Prevalence of energy, (5) Virtualization of energy, and (6) Informationization of energy. Then, the key technologies of EI are presented in three aspects: (1) Physical layer: synergistic energy network; (2) Data and information layer: cyber-physical energy system; and (3) Operation layer: energy operation with an Internet business model. Lastly, representative demonstration projects of EI in China are introduced. We hope the demonstration projects can give readers a specific and vivid perception of the concept, status quo, and prospects of EI in China.

12.1 Concept and characteristics

12.1.1 Concept

With the international energy and economic landscape changing, many countries are advocating energy transition strategies, of which EI is a key representative. In the past 5 years, the concept of EI has thrived in China, and many pilot projects have been or are about to be implemented.

Although different agencies hold different understandings of EI, a representative interpretation should be introduced first. According to the definition of NEA (National Energy Administration of China), the concept of EI (Internet plus smart energy) is as follows: EI is a new form of energy industry development that features the deep integration of the Internet and energy production, transmission, storage, and consumption within an energy market. It has the characteristics of smart equipment, multienergy coordination, distributed supply and consumption, a horizontal system structure, and an open market. China Academy of Sciences' academician Prof. Xiaoxin Zhou points out that EI is a new ecologized energy system that prioritizes renewable energy, uses electric power as its foundation, achieves multienergy type coordination, supply and consumption coordination, and central and distributed production coordination, and advocates mass participation.

12.1.2 Characteristics

The concept of EI in China has six key characteristics.

12.1.2.1 Synergy of energy

Via the technologies of multienergy synergistic operation and dispatch, the operation of multiple energy subsystems, including an electric power grid, thermal/cold network, gas network, petroleum and coal supply and use system, and transportation section can be coordinated to exploit and combine each technology's complementary advantages. The coordination of these different energy types can help them support each other so that wasted energy can be minimized by energy conversion, and peak load can be satisfied with the help of various energies' participation. Thereby the overall efficiency of the energy system can be enhanced, and the use of capital and assets can be improved.

12.1.2.2 Higher energy efficiency

The intrinsic horizontal, marketized feature of the EI system structure values its benefit to society, utility, and effectiveness, interpreting the concept of "energy efficiency" in a broader sense. The benefit includes not only economic but also sociological considerations; therefore, the development of environmentally friendly cost-competitive renewable energies, such as wind and solar power technologies, is advocated to guarantee a comprehensive environmental and sociological benefit for the whole society. The utility means that the whole system should run to satisfy the needs of all participants, including producers, consumers, operators, and supervisors. The efficiency vision stresses the use of technology and operation systems to continue improving the overall efficiency of the energy system.

12.1.2.3 Commercialization of energy

Commercialization means that the commodity properties of energy must be resumed and an open energy market should be established. Thereby the vitality of all the participants of the energy market can be activated and a prosperous energy market ecology can be fostered. Novel business models in the energy market can be formed, including e-commerce energy sales, financialization of energy trade, marketization of investment, Internet-based financing, and so forth. New explorations now include searching for new energy consumption models, creating the atmosphere of an energy sharing economy, and promoting the free trade of energy while building an energy consumption ecological system.

12.1.2.4 Prevalence of energy

All segments of the energy industry should be ubiquitous in society, meaning wider and more flexible and frequent participation of all parties. Energy production will shift from centralized generation to distributed production to achieve this vision of prevalence, in which energy units could be plugged-in and connected with equality, and all energy equipment and terminals could perform two-way communication and

support smart supervision and control. Therefore, all participants of the energy chain can share resources, cooperate with each other, and convert the frontier technologies and innovations into industrial practice. This will create an open innovation system to boost transregional, -field achievement transfer and synergistic innovation.

12.1.2.5 Virtualization of energy

The successful virtualization technology of the Internet can be used as a reference. Software can be used to integrate the distributed infrastructure of an energy system into a virtual resource and to satisfy the flexible demand of the energy system. For example, this could be carried out by cloud commanding the distributed inventory lead acid batteries to uniformly perform the peak loading shaving function for the grid. A virtual power plant is also a typical representation of such an effort. The adoption of the virtualization of energy can dramatically enhance the overall utilization efficiency of the energy and infrastructure of the whole system.

12.1.2.6 Informationization of energy

Informationization means first discretizing energy at the physics level. It then endows information properties on energy via calculation to turn energy into resources that are easy to manage and control, like calculation capacities, band resources, and storage capabilities in the communication field. Thereby future personalized customization of energy operation services can be achieved.

12.2 Key technologies

EI consists of three layers: (1) Physical layer, (2) Data and information layer, and (3) Operation layer. In this section, we will introduce the key technologies of each layer.

12.2.1 Physical layer: synergistic energy network

Coordinated multienergy synergistic networks build up the physical layer and lay the foundation of EI. This layer is the backbone of the system, and the successful operation of the other layers relies on it. It consists of the following key technologies:

12.2.1.1 Energy conversion and multienergy interconnection

The integration of various types of energy allows us to run the whole energy system in an optimized manner. For example, representative energy conversion technologies, such as energy storage technologies, can convert temporarily superfluous heat/electric power into chemical/mechanical energy and deliver it when the system is short on energy; thus the system can balance energy production and demand in a proactive way. Other conversion technologies, such as combined cooling heating and power and power to gas, can also be employed in a similar manner to optimize the system energy

flow. Therefore, these technologies enable us to operate several energy networks in a coordinated way (such as electric, cold, heat, and gas), allowing us to design an optimized overall system operation strategy, and achieve the goals of improving energy efficiency and peak load shaving via the support of a multienergy network.

12.2.1.2 Innovation of DC technology

DC has a long history, but only in recent years has it regained attention and experienced faster development. Line-Commutated Converter—High-Voltage DC is the conventional DC technology that is widely used; however, it requires the AC system to provide commutation current, and commutation failures often occur when connected to a weak AC grid. It also features high reactive power loss and high land occupation. In recent years, advances in semiconductor and power electronics have led to an innovative commutation component, achieving voltage sourced converter high voltage direct current technology. This technology exhibits advantages such as: (1) independent control of active and reactive power and easy to achieve power flow reversal, (2) reactive power support for an AC system to increase its voltage stability, (3) power for a sourceless grid, and (4) a converter station that occupies a small area. The advances of DC also enable more friendly access to renewable energy, electric vehicles, and energy storage; therefore, it is regarded as the evolution trend of the grid system and a key element in supporting EI.

12.2.1.3 Novel power electronics technologies

With design and fabrication techniques approaching a mature level, silicon-based electronic devices are approaching their theoretical limit. Therefore, in recent years, power technologies based on novel materials with a larger band gap, such as SiC and GaN, are developing fast. This material revolution has resulted in the innovation of power electronic devices with enhanced switch properties, such as MOSFET, IECT, IGBT, IGCT, etc. Compared with conventional silicon-based devices, these technologies feature higher working voltage and current, smaller volumes, lower switch loss, higher work frequency, and so on. The typical application of power electronic devices mainly includes solid state transformers and DC breakers, which are also key technologies for DC. A solid transformer can serve as the energy hub of the DC system and coordinate the energy flow, whereas a DC breaker can help the system isolate failure.

12.2.1.4 Renewable energy technologies

Renewable energy is the energy that is collected from renewable resources, which are naturally replenished on a human timescale, such as sunlight, wind, tides, waves, biomass, geothermal heat, and so forth. Renewable energy can be utilized for electricity generation, cold/heat production, and other services. It features advantages such as low or no carbon emission, less pollution during energy production, type diversity, and a high total amount.

The successful promotion of the EI requires a higher amount of renewable energy in the energy mix. In recent years, the fraction of renewable energy is increasing dramatically worldwide, thanks to the enhanced overall performance and cost competitiveness of renewable energy technologies as a result of increased R&D investment and public attention. The use of wind power and photovoltaic technologies is a key representative of the better use of renewable energy. However, the further promotion of renewable energy still faces intrinsic problems, including spatial and time distribution inhomogeneity, intermittency, and low intensity. These features require us to focus more on R&D work and improve other related technologies that can promote the development of renewable energy technologies, such as energy storage, DC, and power electronics technologies.

12.2.1.5 Energy storage technologies

Energy storage is the capturing of energy produced at one time for use at a later time. Energy storage technologies mainly consist of four types: electrochemical, mechanical, electromagnetic, and thermal. Conventionally, the conventional grid system must achieve a real-time balance of generation and consumption, and in most cases, the power flow is unidirectional. However, energy storage technologies enable energy to be stored and converted in another form, and the stored energy can be fed back into the system when needed. Therefore, the introduction of energy storage can change the real-time balance of generation and consumption, providing two-way energy exchange when needed, thereby adding more flexibility and stability to the power system. Energy storage can be deployed in various positions of the grid system to provide versatile services. It can also enable the electricity grid to work in a coordinated way with cold, thermal, gas, and transportation networks.

For EI, the use of energy storage can help the system achieve peak load shaving and deal with the intermittency problem of a high proportion of renewable energy. It can also provide various services, such as UPS, black start, frequency regulation, power quality management, etc., to improve the flexibility and stability of the system. The deployment of energy storage also helps diversify the EI ecology. For example, roof-top photovoltaic users can use energy storage to better balance their energy generation and consumption and adopt a more advantageous strategy when interacting with other users of the system.

12.2.1.6 Energy Routers

The power flow direction in the conventional power grid is often unidirectional due to the centralized, hierarchical structure of the system. However, the conventional power dispatch mechanism commanded by the control center cannot deal with the increased complexity of the energy system with the incorporation of more renewable energy, electric vehicles, and active participation of prosumers. The evolution toward EI advocates peer-to-peer energy trade/exchange between participants, leading to a complex power flow feature of the whole system. As a result, there must be a key “hub” component that can manage access and the control of power flow between different participants in the network. This vital device is called the Energy Router.

The Energy Router can be developed by referring to the hub devices of the Internet: it should provide access for energy generation, consumption, and transmission infrastructures. It should also have the functions of connecting different energy units (distributed energy generation units, microgrid units, and so on), supervision and dispatch of energy, communication, and maintenance. Furthermore, it is obvious that the Energy Router must coordinate with all three layers of the EI, representing the key spirit of EI: “merging energy and information infrastructures.”

12.2.1.7 Wireless electric power transfer

Wireless electric power transfer is the transmission of electrical energy from a power source to an electrical load, such as an electrical power grid or appliance, without the use of conductors such as wires or cables. It depends on time-varying electric, magnetic, or electromagnetic fields to transmit field energy across an intervening space to one or more receivers, where it is converted back to an electric current and then used. Wireless power uses the same fields and waves as wireless communication devices such as radios, and its main technologies include inductive coupling, resonant inductive coupling, capacitive coupling, and magnetodynamic coupling. Wireless electric power transfer can make the charging and energy transfer process much more convenient and smart. First, it can greatly improve the charge experience when we use battery-supported devices, such as cell phones and electric vehicles. What is more, as there would be no need to set up wires and plug-in such devices, the (dis)charge and energy transmission action can be done automatically by the energy system. This characteristic means better utilization and management of all energy terminals of EI.

Currently, wireless electric power transfer is still facing problems, such as low transmission efficiency, high cost, and short distance, thus requiring more R&D to further improve the performance of this technology.

12.2.2 Data and information layer: cyber-physical energy system

The data and information layer is built above the physical layer. It collects data from the physical layer, analyzing and transmitting valuable energy information to the operation layer to coordinate the system operation. Its key feature is the integration of energy and information.

12.2.2.1 Smart energy terminals and data collection

The vision of EI shares some features with the Internet of Things. One key characteristic is that they both adopt smart terminals to connect and integrate all the participating elements in the physical layer into the network. The smart energy terminals are multifunctional: (1) they can integrate a multienergy-type monitoring function into one terminal, thereby assisting multienergy synergy and optimizing system energy flow and efficiency; (2) they can collect terminal data of various types at a high frequency and send them to the data and information layer and operation layer to provide basic information necessary for system operation; (3) they can also guide

and control the user's energy use behavior, acting as the essential element for various energy services, such as demand response and so forth.

12.2.2.2 Advanced communication technologies

The successful operation of EI means the merge of energy and information. The two-way exchange of information between energy terminals and the information/operation layer leads to a drastically increased amount of information that needs to be transmitted and processed. This workload requires advanced communication technologies to be applied for EI. Key requirements for advanced communication technologies are: (1) collecting information at a high quality, (2) sending information at a high speed, (3) processing large amounts of information with high efficiency, and (4) ensuring system information safety.

12.2.2.3 Energy management system

The introduction of various renewable resources and other flexible loads adds more uncertainty and intermittency to the energy system; therefore, EI must use the energy management system to control and coordinate the operation of energy production and consumption, optimize the coordination of different energy types, and improve interaction between different consumers/prosumers/producers, thereby also improving the energy efficiency and economic value of the whole system. The operation of an energy management system can follow a central pattern or distributed microgrid-oriented style, depending on the structure, market mechanism, conditions, and will of the producers/consumers of the system. Such a system can send commands to the physical layer and cooperate with the operation layer; making it is a key element that could connect the three layers of EI together.

12.2.2.4 Big data

The conventional energy system is generally closed and energy information is often underutilized. This problem has, in part, led to the unsatisfactory efficiency of energy systems such as the grid. The fusion of energy and information in EI supports better use of energy information to help boost development, optimize energy dispatch, and improve the operation of the energy system. Big data in the energy area/EI area covers not only information of various energy types (petroleum, grid, gas) but also energy-related socioeconomic information such as population, geography, and meteorology. It also covers areas such as data collection, processing, analysis, and application. Furthermore, energy big data is the basic information source supporting various business models and can help elucidate the operation condition of an energy system and specific needs of each participant, so that a suitable customized business model can be accordingly established.

12.2.2.5 Cloud computing

The successful utilization of big data in the energy field needs cloud computing to provide deeper and further value excavation, turning information into wealth. The cloud computing concept in the Internet industry advocates for the connection and sharing of

computing power for any participant in urgent need. The future energy system will have a larger fraction of distributed energy resources and much more diversified interaction between different participants, including more frequent information production and exchange; therefore, data processing ability, or computing resources, are in urgent need, and the optimized integration of all accessible computing resources of the energy network into the cloud is definitely a route toward the solution.

12.2.2.6 Blockchain technologies

Blockchain is a distributed database that maintains a continuously growing list of records, called blocks, secured from tampering and revision. Such technology uses a peer-to-peer network and a distributed timestamping server; therefore, the database is supported by distributed participants and runs in an autonomous pattern. Each block contains a timestamp and a link to the previous block, and its modification needs approval from more than half of the participants of the system. Consequently, the information is resistant to malicious modification, and all participants can help maintain the credibility of information in the system.

The value of blockchain technology is that it advocates for all individuals' participation and work in a decentralized manner, thus adding more flexibility compared with conventional vertical hierarchical information structures, favoring much more frequent information exchange with high credibility and efficiency. As a result, it is very useful for helping future EI information systems work in a more diversified, noncentralized way.

12.2.3 Operation layer: energy operation with an Internet business model

The operation layer of EI features an innovative operation style of the energy system. It uses the information collected from the data layer and devises different energy business patterns to guide the operation of the physical layer. Devising a business model and supervising market operation is the main function of the operation layer. It consists of the following key technologies.

12.2.3.1 Energy trade platform

The development of EI needs the establishment of a novel market mechanism, meaning energy trade platform technology is needed to support it. Such a platform should be designed based on the following main considerations: (1) promote marketization of energy to resume its property as a commodity, (2) support the role transformation of conventional grid enterprises into energy service providers, (3) provide an open and fair area for energy service to be introduced and traded. The ultimate goal of the energy trade platform is to introduce the power of a market into the energy system to coordinate energy production, transmission, and production according to real need and enhance new energy access and utilization.

12.2.3.2 User interaction platform

The EI system is decentralized, market-oriented, and advocates for wide, active participation of all users in the system to prosper the energy market. The introduction of a vast number of distributed energy sources/owners and their participation in the energy market requires an area open and fair for user interaction. It can give different users opportunities to learn energy information, publish information, and make preliminary contacts essential for energy trade.

12.2.3.3 Real-time energy trade and clearance technologies

The operation of EI needs to deal with frequent interaction between different users. Because energy production and consumption are generally real-time based and highly sensitive to various conditions (e.g., renewable energy can be intermittent), technologies for real-time energy trade and clearance are needed to support business activities on the platform and must be implemented in an efficient way.

12.2.3.4 Energy Internet financial technologies

The fusion of financial and Internet technologies has resulted in a revolution in the financial industry, making financial technology highly needed for EI. First, the establishment of EI needs a huge investment, in which the financial industry could find an opportunity. Second, EI will also activate the energy markets and produce different, new market operation strategies in which financial services can find a chance. Third, the development of EI will lead to a greater energy industry with a larger volume, as well as integrated interconnection between different industries. Therefore there is room for profit, which may attract the financial industry to join.

12.2.3.5 Value-added energy service

Typical EI ecologies include multienergy synergy, optimized energy production and consumption, and specialized and customized energy service for consumers. In such systems, personalized, individualized energy service needs can be analyzed and solved. In EI, an energy service can be delivered to solve specific energy problems, thus it can be designed as a special value-added energy service to satisfy needs of consumers.

12.2.3.6 Virtual power plant

Virtual power plant (VPP) technology can assemble all controllable distributed energy sources of the energy network to collaborate, integrating their scattered capabilities to serve certain needs of EI. It uses key technologies such as communication, smart metering, data processing, and so forth and could be a framework technology for incorporating various scattered energy sources, such as renewable energy, on a large scale into one powerful source contributing to system operation. The participation of VPP can be a main contribution to various energy service markets, such as ancillary services, carbon trade, and so on. It can also dramatically improve the utilization rate of distributed energy resources, making the whole system work in a more efficient, economically profitable way.

12.3 Representative demonstration projects

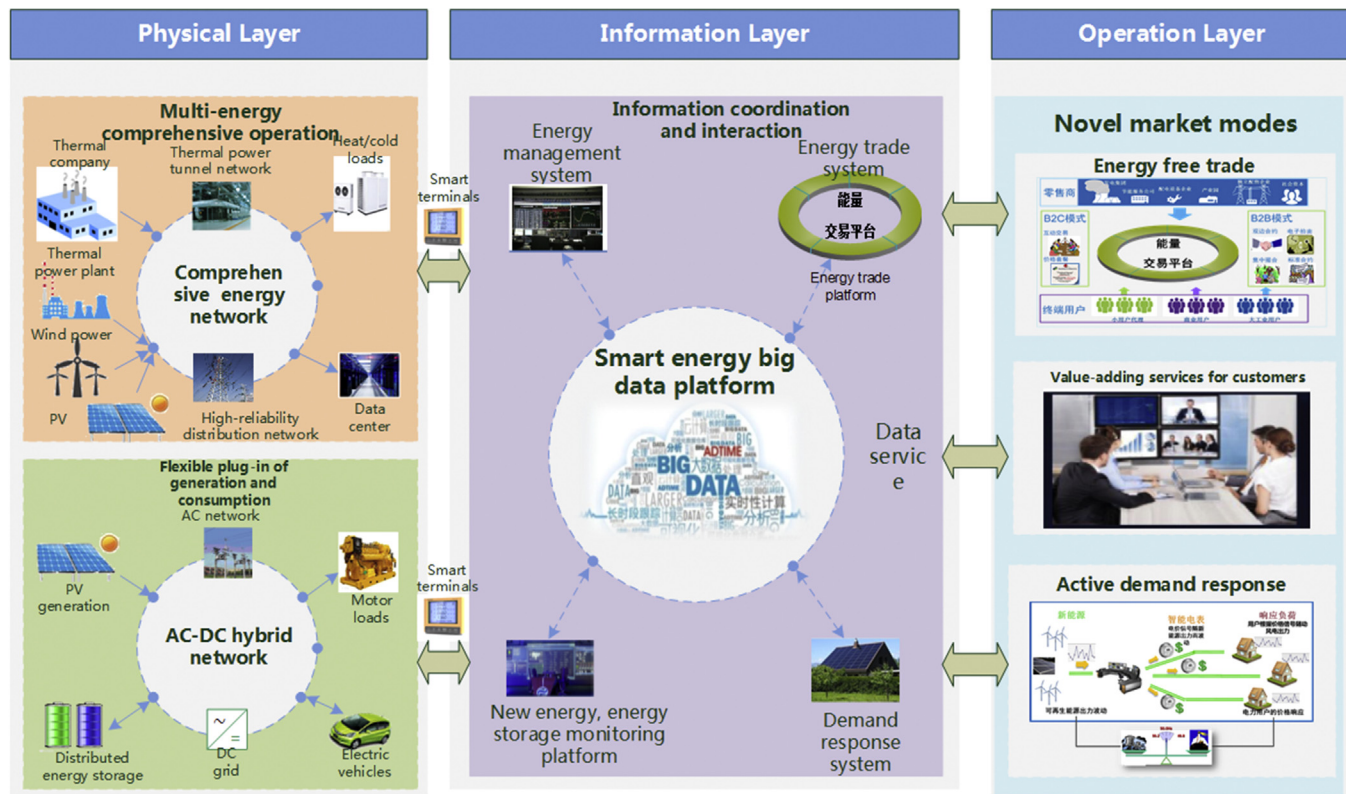
Since 2015, projects featuring EI have begun to be built throughout China. These projects all exhibit the key characteristics mentioned above and also have special features of their own. Because of the need for energy transition and energy system revolution, as well as the search for a method to boost economic growth, the successful operation of these pilot projects of EI has attracted the attention of the country. The Chinese government has issued several notices on EI since 2015. In 2016, NDRC (National Development and Reform Commission) issued the guideline “NDRC-Energy [2016] 392,” aiming to promote the development of “Internet +” smart energy. In the same year, the NEA issued the notice [2016]200, aiming to boost the application and implementation of “Internet +” smart energy (EI) demonstration projects. The notice received an enthusiastic response from society and more than 300 applications were received. After a careful screening and open reply process, 55 projects were approved. These projects cover almost all areas of society, encouraging exploration and integration of EI in agriculture, industry, transportation, commerce, sports, education, and so forth.

Because of space limitations, we selected representative examples for our readers.

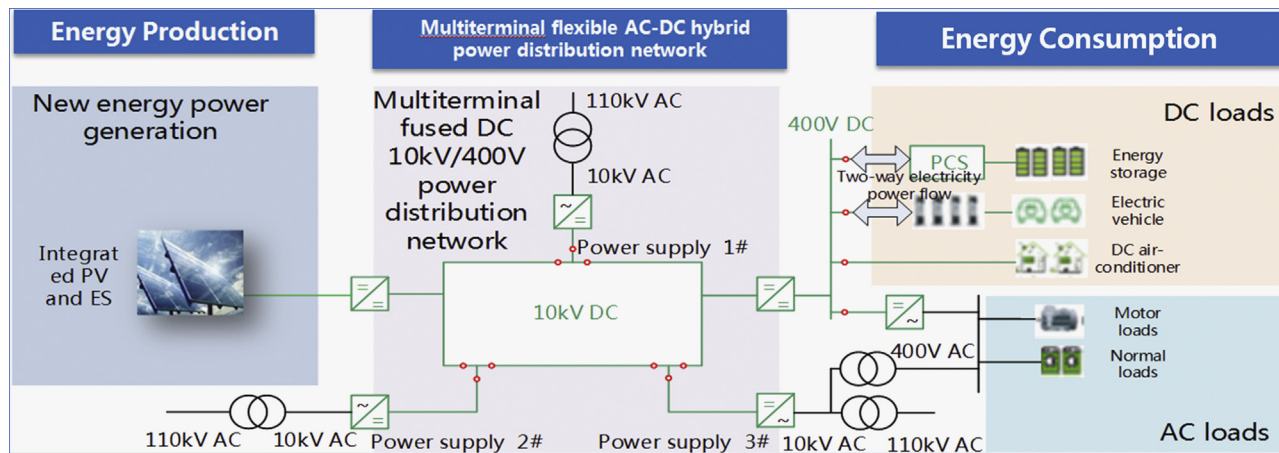
12.3.1 *Optimized multienergy coordination and novel comprehensive energy system in Zhuhai, Guangdong province*

Guangdong Grid Co., Ltd., proposed the “Internet +” smart energy demonstration program “City-park double level smart energy project supporting energy consumption revolution.” The project will be built in Hengqin free trade zone and Tangjia Science Park of the City, and the two parts will coordinate so that a city-park double level energy system will be established. The project will build a high-efficiency green comprehensive energy system (physical layer), an advanced energy information system (information layer), and an innovative energy service consumption model (operation layer). The project aims to increase energy efficiency on the user side by 5%–10%, decrease the peak-valley difference by 10%, enhance the consumption of new energy, and facilitate the deployment of new loads, including electric vehicles.

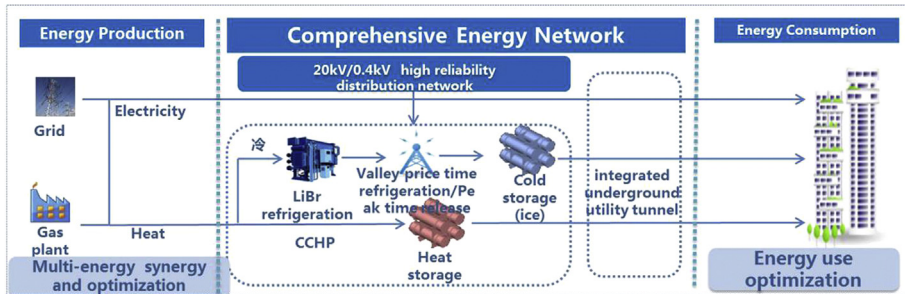
The project will build a multienergy synergistic energy network in Hengqin free trade zone, which utilizes advantages such as the local Wangyang thermal plant to produce electricity and act as a cold source, as well as the local integrated underground utility tunnel for the integrated transmission of various energy resources. A novel ± 10 kV/400 V flexible DC grid will be built in the Tangjia region, and it can support flexible plug-in of various novel loads with high accessibility, efficiency, and reliability, dramatically increasing the robustness of the grid, satisfying the need for high-quality power service of the hi-tech industry and enhancing utilization of new energy in the area.



General framework of Zhuhai-based EI project.



10 kV/400 V flexible DC grid in the Tangjia region.



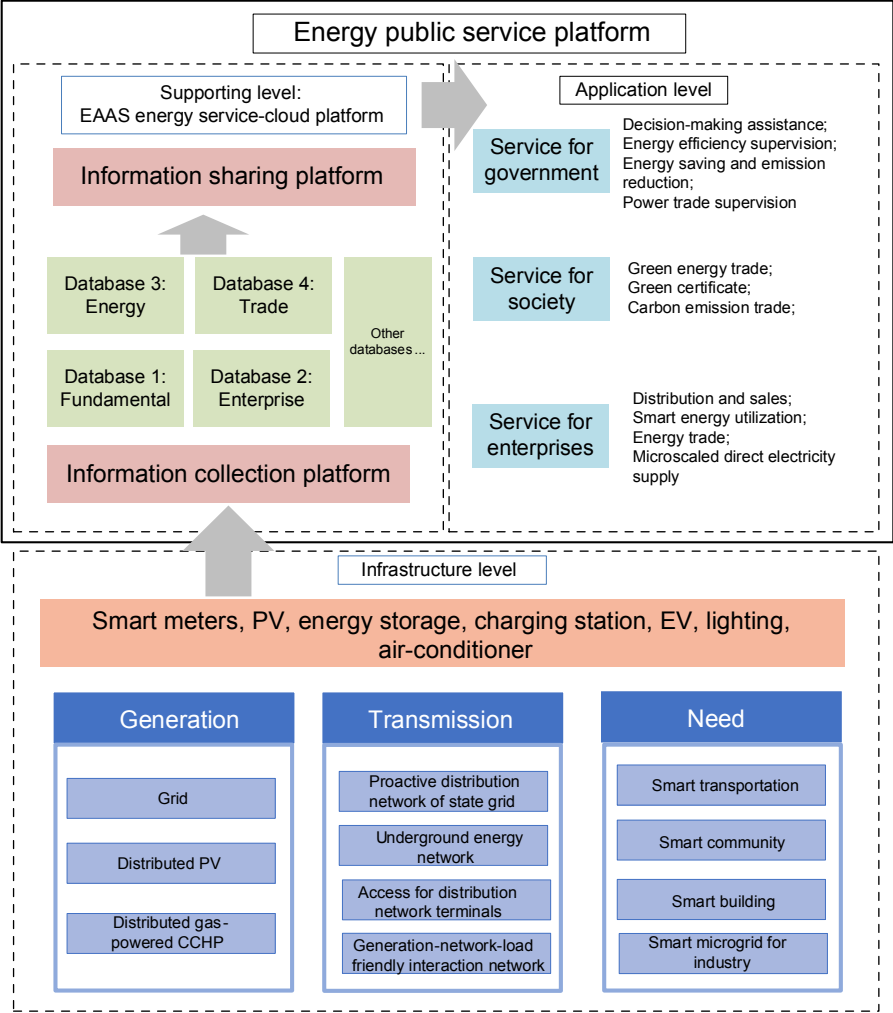
Multienergy interconnected energy network in Hengqin free trade zone.

What is more, an advanced information terminal network, information big data platform, and cloud computing ability will also be developed to analyze local energy data, supporting an implementation that attempts energy marketization. Novel energy business models will be established, including a proactive demand response system, smart energy management platform, which support versatile energy value-added services, and a trade platform, which encourages more participation of energy prosumers from different parts of the city. The result is achieving the goal of “promoting energy saving and CO₂ emission reduction, increasing energy utilization efficiency, lowering the energy cost, and creating an innovative energy service model.”

12.3.2 Multipart participation and energy information platform coordinated interaction

The Suzhou Industrial Park Energy Internet Project is proposed by the Suzhou Industrial Park Administrative Committee. Suzhou Industrial Park is located in the Yangtze Delta region, a most prosperous part of China. This park has many enterprises and its annual GDP has reached 206 billion rmb. The current energy status of this park is: (1) high energy consumption, (2) high energy management level of settled enterprises, (3) aggregated energy service industry, (4) high informationization level of energy.

To achieve energy innovation and increase the holistic competitiveness of the area, Suzhou Industrial Park has applied the EI demonstration program, aiming to achieve the following goals: (1) increase penetration of distributed PVs and their share of green power consumption; (2) build a regional multienergy interconnection network; (3) achieve information sharing and coordination by using an energy public service platform to balance and optimize energy supply and need; (4) adopting the spirit of EI in various sections of society, including industry, construction, transportation, day-to-day life, and so forth.

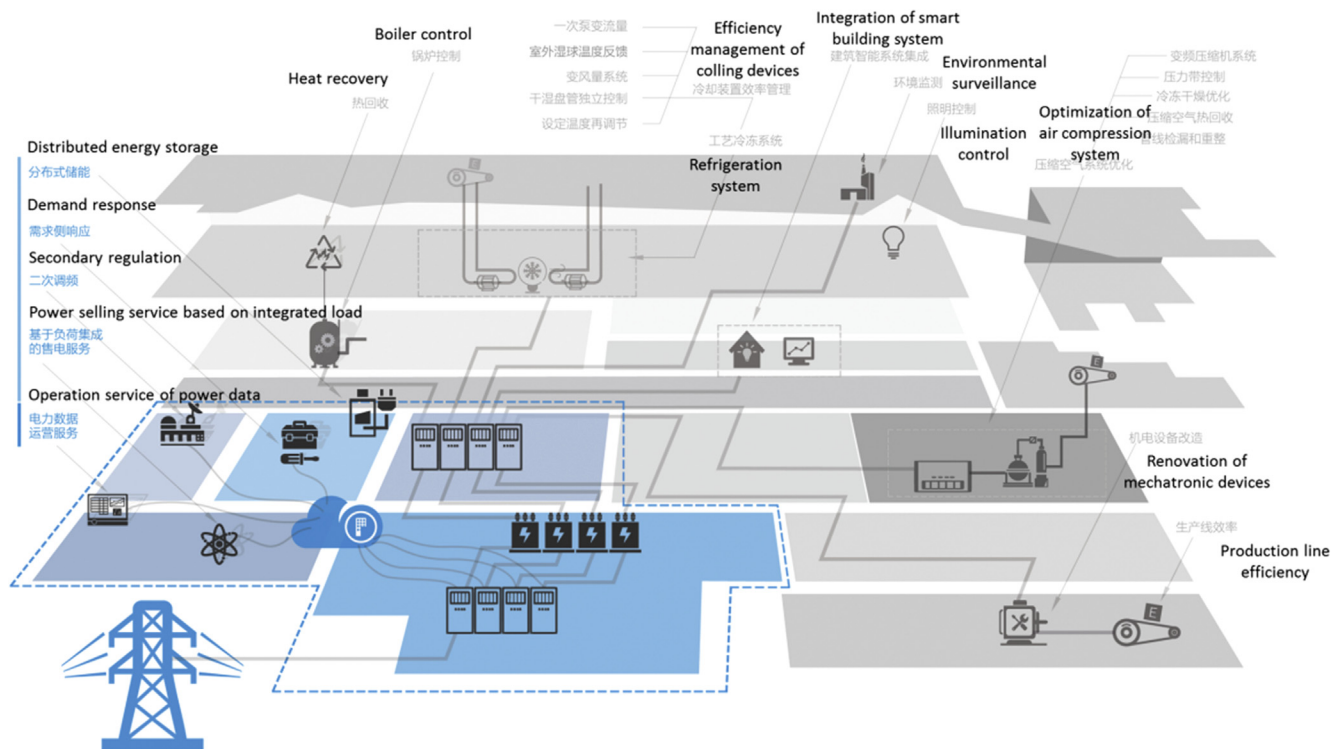


Technical framework of the Suzhou based Energy Internet project.

More than 10 enterprises with strong energy backgrounds and sufficient R&D and engineering capabilities have attended to this project, including State Grid Suzhou Company, GCL Smart Energy Co., Ltd., Zhonhen Electrics Co., Ltd., and others. Together they will build a system of three levels: (1) infrastructure level consisting of coordinated energy generation, transmission, and consumption; (2) supporting level, which will provide overall energy information service to all participants in the energy system; (3) application level, which will provide energy service to the government (energy information, energy efficiency, and trade supervision), society (green energy trade, green certificate, emission trade service), and enterprises (energy retail, smart energy service, and so on). The supporting level and application level combine to form the core part of the project—the energy public service platform. It will promote coordination of the generation-network-load of the park, achieving a balance of energy generation and production and efficient supervision, merging the spirit of EI with the future development of the city.

12.3.3 Energy service on the user's side provided by Shenzhen Cubenergy Co., Ltd.

The current grid system revolution of China advocates deregulation of the power generation and selling sector, and Guangdong province has taken the lead in this exploration and revolution era. In recent years, a lot of enterprises have emerged in Guangdong, focusing on novel business in the energy industry chain. Shenzhen Cubenergy Co., Ltd., is one typical company, concentrating on providing energy solutions for end-users based on EI concepts. Their service is data-oriented, technology-based, and user-demand targeted. They usually first carry out a comprehensive study on industrial-commercial users' power consumption by high-precision and frequency data analysis to find out the energy saving potentials. Afterward, Cubenergy will implement engineering innovations for end-users and offer them customized professional energy service. By doing so, the users' energy consumption can be dramatically cut and their electricity bills will be effectively reduced.



Schematic of Cubenergy's energy service areas.

Cubenergy also forms a win–win cooperation mode with customers: Cubenergy provides distributed energy storage on the users' side, which tries to least affect the original operation of the customers. The energy storage will benefit users by a peak load shaving effect, power quality assurance, and upgrade deferral. In the meantime, Cubenergy will also earn profits due to cloud-based optimized manipulation of all their distributed energy storage devices, done to achieve the best benefit for all energy network participants. The distributed resources can also be commanded to work aggregately to provide more beneficial services for the whole grid, such as frequency regulation, upgrade deferral, and so on. Therefore, Cubenergy adopts one central idea of EI, which is that we can integrate distributed resources to provide energy service for the whole energy network so that a more efficient, reliable, and safer energy environment can be created. Cubenergy has cooperated with many renowned companies in China, including Hisense, Kempinski, P&G, and Skyworth, and successfully removed a considerable amount from customers' energy bills.